Fall Sem. 2024-25

Tutorial– 3

Programme Name & Branch: B. Tech.

Course Name & Code: Complex Variables and Linear Algebra (BMAT201L)

Practice all the questions:

1. Obtain the Taylor's series expansion of the function $f(z)=\dfrac{1}{z^2-z-6}$ about $z=1$.

   Answer: $f(z)=\dfrac{-1}{6}+\dfrac{1}{5}\left(\dfrac{1}{2^2}-\dfrac{1}{3^2}\right)(z-1)-\left(\dfrac{1}{2^3}+\dfrac{1}{3^3}\right)(z-1)^2+\left(\dfrac{1}{2^4}-\dfrac{1}{3^4}\right)(z-1)^3+\ldots$

2. Obtain the Laurent series expansion of the function $f(z)=\dfrac{z}{z^2-4z+3}$ in the region $1<|z|<3$.

   Answer: $f(z)=\dfrac{-1}{6}\left(1+\dfrac{z}{3}+\dfrac{z^2}{3^2}+\ldots\right)-\dfrac{1}{z}\left(1+\dfrac{1}{z}+\dfrac{1}{z^2}+\ldots\right)$

3. Discuss the singularities of (i) $f(z)=\dfrac{1-\cos z}{z}$ (ii) $f(z)=e^{1/z}$

4. Find the poles and the corresponding residues at each pole of the function $f(z)=\dfrac{z^2+2z}{(z+1)^2(z^2+4)}$.

   Answer: Here $z=-1$ is a pole of order two and $z=-2i,\ z=2i$ are simple poles. And the corresponding Residues are $\dfrac{-2}{25}, \dfrac{1+7i}{25}, \dfrac{1-7i}{25}$ respectively.

5. Evaluate $\displaystyle\int_C \frac{z^3e^{-z}}{(z-1)^2}dz$ where C is $|z-1|=\dfrac{1}{2}$.

   Answer: By Cauchy's integral formula, $\displaystyle\int_C \frac{z^3e^{-z}}{(z-1)^2}dz=\frac{4\pi i}{e}$

6. Evaluate $\displaystyle\int_C \frac{z^2-1}{z^2+1}dz$ where C is $|z-i|=1$.

   Answer: By Cauchy's integral formula, $\displaystyle\int_C \frac{z^2-1}{z^2+1}dz=-2\pi.$

7. Evaluate $\displaystyle\int_c \frac{z^2+2z}{(z+1)^2(z^2+4)}dz$ where C is $|z|=\dfrac{3}{2}$ using Residue theorem.

Answer: Here $z=-1$ is a pole of order two which is inside of C and $z=-2i,\ z=2i$ are simple poles which are outside of C. Therefore $\int_c \frac{z^2+2z}{(z+1)^2(z^2+4)}dz = 2\pi i(\operatorname{Re}s.f(z))_{z=-1} = 2\pi i\left(\frac{-2}{25}\right)$

8. Evaluate $\int_C \frac{4-3z}{z(z-1)(z-2)}dz$ where C is $|z|=\frac{3}{2}$ using Residue theorem.

Answer: $\int_C \frac{4-3z}{z(z-1)(z-2)}dz = 2\pi i$.

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Tutorial- 4

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Practice all the questions

1. Evaluate $\int_0^{2\pi} \frac{\cos 2\theta}{5+4\cos\theta}\, d\theta$ using Residue theorem. Answer: $\frac{\pi}{6}$

2. Evaluate $\int_0^{\pi} \frac{1}{1+\sin^2\theta}\, d\theta$ using Residue theorem. Answer: $\frac{\pi}{\sqrt{2}}$

3. Evaluate $\int_0^{\pi} \frac{1+2\cos\theta}{5+4\cos\theta}\, d\theta$ using Residue theorem. Answer: 0

4. Evaluate $\int_0^{\infty} \frac{2x^2-1}{x^4+5x^2+4}\, dx$ using Residue theorem. Answer: $\frac{\pi}{4}$

5. Evaluate $\int_0^{\infty} \frac{x\sin x}{1+x^2}\, dx$ using Residue theorem. Answer: $\frac{\pi i}{2e}$

6. Evaluate $\int_{-\infty}^{\infty} \frac{\cos 2x}{x}\, dx$. Answer: 0

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Tutorial- 5

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Practice all the questions

1. Find the eigenvalues and the corresponding eigenvectors of $A = \begin{pmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{pmatrix}$

   Answer: eigenvalues are $\lambda = 0, 3, 15$ and the corresponding eigenvectors are

   $X_1 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$, $X_2 = \begin{pmatrix} -4 \\ -2 \\ 4 \end{pmatrix}$, $X_3 = \begin{pmatrix} -2 \\ 2 \\ -1 \end{pmatrix}$.

2. Verify Cayley-Hamilton theorem for $A = \begin{pmatrix} 2 & 1 & 2 \\ 5 & 3 & 3 \\ -1 & 0 & -2 \end{pmatrix}$ and hence find A^{-1} if exists. Also find the matrix B where $B = A^5 - 3A^4 - 7A^3 - A^2 + 2A + \mathrm{I}$

Answer: Characteristic equation of A is $\lambda^3 - 3\lambda^2 - 7\lambda - 1 = 0$. Clearly, $\mathrm{A}^3 - 3\mathrm{A}^2 - 7\mathrm{A} - \mathrm{I} = 0$ and $\mathrm{A}^{-1} = \begin{pmatrix} -6 & 2 & -3 \\ 7 & -2 & 4 \\ 3 & -1 & 1 \end{pmatrix}$. And $B = 2A + \mathrm{I}$.

3. Solve $x + 2y - 5z = -2$, $2x - y + 3z = 4$, $x + 3y - z = 3$ using (i) Gaussian elimination and (ii) Gauss Jordan and Gauss Jordan methods.

   Answer: x=1, y=1, z=1

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Tutorial- 6

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Practice all the questions

1. Let $R^4(R)$ be a vector space and $W=\{(x, y, z, w) \mid x+y=0, z=2w\}$ a sub set of R^4. Verify whether W is a subspace of $R^4(R)$ or not?

Answer: W is a subspace of $R^4(R)$.

2. Is the set $S=\{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$ linearly independent?

Answer: The set $S=\{(1, 2, 1), (3, 1, 5), (3, -4, 7)\}$ is linearly dependent

3. Is the set $S=\{(2, 1, 4), (1, -1, 2), (3, 1, -2)\}$ forms a basis of a vector space $R^3(R)$.

Answer: Yes, S is a basis of $R^3(R)$.